AI-POWERED SECURITY SYSTEM · EMBEDDED + CLOUD

SMART X-RAY LUGGAGE DETECTION SYSTEM

Real-Time Dangerous Object Detection · Automated Hardware Response via STM32

Harshal Dhavan · Chirag Pingale · Yash Fanse · Laxman Birajdar
Department of Electronics and Telecommunication
Rajiv Gandhi Institute of Technology, Andheri

OBJECTIVE 01
Detect 5 threat classes in baggage using a custom-trained YOLOv8 model at 25fps

OBJECTIVE 02
Automatically stop conveyor belt & actuate solenoid via STM32 over UART serial

OBJECTIVE 03
Provide 3-mode state machine — Safe / Danger / Emergency — with physical LED & LCD feedback

OBJECTIVE 04
Sync all detection events with snapshots to a globally accessible cloud dashboard via WebSocket

YOLOv8	5	25fps	≥80%	CUDA	9600
AI Model	Threat Classes	Live Inference	Conf. Threshold	GPU Accelerated	UART Baud Rate

SYSTEM ARCHITECTURE

INPUT | USB Camera (1280×720 · 30fps) → OpenCV Frame (Capture · BGR) → Pre-process (Brightness · Flip · Rotate)

AI ENGINE | YOLOv8 Model (best.pt · CUDA GPU) → Bounding Box (Annotated Frame) → FSM Confirm (12 frames · deque)

DECISION | SAFE → "0" (Flask · Serial) + DANGER → "1" (Flask · Serial) → UART COM5 (9600 baud · PC→STM32)

HARDWARE | STM32F4 (PB10·PB12·PA0-7·PB0-1) → Belt Motor (Relay NC · PA4) + Solenoid (Relay NO · PA5) + Buzzer+LEDs (PA0 · PB0·PB1·PA7) + 16×2 LCD (I²C · 0x27)

CLOUD | WebSocket (wss:// · Railway) → FastAPI Backend (Railway · async) → PostgreSQL (Detection logs) → Admin Panel (JWT · Web + Mobile)

VISUAL REPRESENTATION

Mains 230V AC Input → SMPS 12V / 10A DC Output

- Buck Conv. 1 — 5V / 3A → STM32 (Clean Logic)
- Buck Conv. 2 — 5V / 3A → Sensors & Components
- 12V Direct — Belt Motor — NC Relay · PA4
- 12V Direct — Solenoid — NO Relay · PA5

STM32 PIN REFERENCE

OUTPUTS (DO / PWM)	INPUTS / COMMS
PA4 — Belt Relay (NC)	PB12 — Toggle Switch
PA5 — Solenoid (NO)	PB10 — E-Stop Button
PB1 — Red LED	UART — RX/TX Serial
PA7 — Green LED	I²C — 0x27 LCD 16×2
PB0 — Yellow LED	DPST — Motor Direction
PA0 — Buzzer (tone)	PWM — Motor Speed

OPERATING MODES — STATE MACHINE

SAFE MODE

Conveyor Belt	RUNNING
Solenoid (PA5)	OFF
Buzzer (PA0)	OFF
LCD Display	STATUS SAFE
LED Status	GRN RED YLW

UART: PC sends "0" → motorOn() · solenoidOff()

DANGER MODE

Conveyor Belt	STOPPED
Solenoid FSM	EJECT CYCLE
Buzzer (PA0)	1kHz ALARM
LCD Display	STATUS: DANGER
LED Status	GRN RED YLW

UART: PC sends "1" → motorOff() · SOL_WAITING→EXTENDED

EMERGENCY STOP

Conveyor Belt	STOPPED
Solenoid (PA5)	OFF · SAFE
Buzzer (PA0)	1kHz ALARM
LCD Display	EMERGENCY STOP
LED Status	GRN RED YLW

Hardware: PB10 E-Stop Btn → 50ms debounce → activateEStop()

AI DETECTION PIPELINE

01	02	03	04	05	06	07	08	09
Camera Input	Pre-Process	YOLOv8 Inference	FSM Confirm	UART Command	Hardware Response	Snapshot Save	Cloud Sync	Admin Dashboard
USB · 1280×720 cv2.VideoCapture	Brightness · Contrast Flip · Rotate · Sharpen	best.pt · CUDA GPU conf ≥ 0.80	deque · 12 frames consecutive check	COM5 · 9600 baud "0"=SAFE · "1"=DANGER	Belt stop · Solenoid Buzzer · LEDs · LCD	JPEG · SQLite local + cloud push	WebSocket · FastAPI PostgreSQL · Railway	JWT Auth · Web Live feed + logs

DETECTABLE THREAT CLASSES

Gun · Knife · Scissors · Pliers · Wrench

KEY FEATURES

- Real-time GPU inference at 25fps via CUDA
- 12-frame FSM confirmation — eliminates false positives
- Detection snapshots saved locally (SQLite) & to cloud
- Globally accessible admin panel via JWT-secured login
- Solenoid eject FSM — auto retract after 1s extension
- Hardware emergency stop with 50ms debounce (PB10)
- Live MJPEG + WebSocket stream to web & mobile
- Remote settings sync — conf threshold, frame count, camera
- 3 physical LEDs + LCD display real-time mode feedback
- pywebview desktop app + phone access on local network
- CSV + HTML report export with detection snapshots
- Multi-camera support with auto-failover switching

SYSTEM DESCRIPTION

A full-stack **embedded + AI security platform** for automated baggage scanning. A GPU-accelerated Windows PC runs a custom-trained **YOLOv8** model (best.pt) via a **Flask + pywebview** desktop application, capturing live frames using **OpenCV** and applying an image processing pipeline before inference. Threats confirmed across **12 consecutive frames** (FSM) trigger a UART command ("1" or "0") to the **STM32F4**, which runs a 4-state solenoid FSM to eject the bag, stops the NC relay belt motor, sounds a **1kHz buzzer pattern**, updates 3 RGB LEDs and a **16×2 I²C LCD**. A custom PCB uses **230V SMPS → dual 5V bucks** for clean logic power, with 12V direct to motor and solenoid via relays. Simultaneously, all events and JPEG snapshots are pushed to a **FastAPI + PostgreSQL** backend on Railway via **WebSocket**, accessible globally through a JWT-secured admin panel on web and mobile.

SOFTWARE STACK

AI MODEL	LOCAL BACKEND	DESKTOP UI	CLOUD API	DATABASE	LOCAL DB	VISION
YOLOv8	Flask	pywebview	FastAPI	PostgreSQL	SQLite	OpenCV

REALTIME	AUTH	PYTHON	DEPLOY	GPU
WebSocket	JWT	Anaconda	Railway	CUDA